

Advanced (Carolina Reaper)

Q1. What is the vertex of the graph of $y = |x - 3| + 2$?

- (A) (3, 2)
 (B) (2, 3)
 (C) (3, -2)
 (D) (-3, 2)
 (E) (0, 3)

Q2. The graph of $y = |2x - 1|$ has an axis of symmetry of

- (A) $x = -1$
 (B) $x =$

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- (B) $x = 0$
 (C) $x = 1$
 (D) $x = 2$

Q3. Sarah spends $|20 - t| + 10$ dollars on hobbies t hours after waking up. What is the minimum amount she spends?

- (A) 10
 (B) 20
 (C) 30
 (D) 40
 (E) 50

Q4. The graph of $y = -|x - 2| + 1$ opens

- (A) upward
 (B) downward
 (C) leftward
 (D) rightward
 (E) not at all

Q5. A store has a $|p - 100| + 50$ dollar discount on an item. What price p gives the maximum discount?

- (A) 50
 (B) 75
 (C) 100
 (D) 125
 (E) 150

Q6. The equation $y = |x + 2| - 3$ has a minimum value of

- (A) -5
 (B) -3
 (C) 0
 (D) 2
 (E) 3

Q7. The graph of $y = 2|x| - 1$ has a vertex at

- (A) (0, -1)
 (B) (1, 1)
 (C) (0, 0)
 (D) (0, 1)
 (E) (2, 3)

Q8. A time management app charges $|t - 30| + 10$ dollars for t minutes of usage. What is the minimum charge?

- (A) 10
 (B) 20
 (C) 30
 (D) 40
 (E) 50

Open Ended Questions (FRQ)**Q9.** Graph the function $y = |x + 1| - 2$ and identify its vertex, axis of symmetry, and direction of opening. Provide your answer in the space below.**Q10.** A shopping mall has a parking fee of $|t - 2| + 5$ dollars for t hours. Determine the minimum fee and the time at which it occurs. Show your work and provide your answer in the space below.

Chef's Tips

- Tip 1: Ensure students have graph paper for questions 9 and 10
- Tip 2: Monitor students for any signs of struggle with transformations
- Tip 3: Encourage students to use real-life examples for word problems